Assignment 02

Course No: MAT110 **Semester: Fall 2025** **Total Marks: 15**
Last Day of Submission: 23rd December, 2025.

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1. Compute partial derivative: (2×2)
   a. Let $f(x, y, z) = \sqrt{xy} + \ln(x^2 z^3) - x \tan z$. Find $f_x, f_y, f_z, f_{xz}, f_{xyz}$.
   b. Let $f(x, y, z) = (x^2 - y^2) \cos x + x^5 z^2 - 30z$. Find f_{xy}, f_{xxz}, f_{zyx}.
2. Use an appropriate form of the chain rule: (2×2)
   a. If $z = \ln(2x^2 + y)$; $x = \sqrt{t}, y = t^{\frac{2}{3}}$, find $\frac{dz}{dt}$.
   b. If $w = x^2 + y^2 - z^2$; $x = \rho \sin\varphi \cos\theta$, $y = \rho \sin\varphi \sin\theta$, $z = \rho \cos\varphi$, find $\frac{\partial w}{\partial \rho}, \frac{\partial w}{\partial \theta}$.
3. Locate all relative maxima, relative minima, and saddle points, if any: (2×3.5)
   a. $f(x, y) = x^2 + y^2 + \frac{32}{xy}$
   b. $f(x, y) = x^2 + 2x + y - e^y$